



Attention Kernels and Feature Interference Under a Representational Bottleneck

Anirudh P

2026

Abstract

The arrival of AI promised to revolutionise all industries, but in high-stakes fields like medicine, law and aerospace, its deployment is restricted by a lack of real-time safety verification. An obstruction to this is superposition, where a model represents more features than it has dimensions available, causing otherwise distinct features to share representational directions, tangling (increasing interference in) the model’s internal logic. Existing approaches primarily address this problem after training (post-hoc) particularly through Sparse Autoencoders, whereas architectural interventions aim to prevent undesirable representations from forming in the first place. This project investigates whether the attention kernel (the function that converts similarity scores into attention weights) can act as an architectural control, allowing for untangled internal logic, and models to be verified in real-time.

I compared softmax attention with a bounded sigmoid and squared-ReLU alternative using synthetic data with known ground-truth features. My experiment used a factorial design (kernel $\times$ bottleneck) with 8 and 20-dimensional representations, allowing me to investigate the effect of kernel choice and compression pressure. I measured feature interference, feature recoverability, reconstruction performance, attention concentration as well as a random direction chance-level baseline for comparison. A follow-up experiment used a temperature of 0.05 to force a higher attention concentration.

I found that the bounded kernel did not meaningfully reduce interference relative to softmax because softmax attention remained close to uniform, leaving little concentration for the bounding mechanism to constrain. Even when concentration was encouraged in the follow-up, the network only partially adopted the imposed behaviour, suggesting that learned query and key weights can compensate for architectural constraints during training. In contrast, the hard-sparse squared-ReLU kernel consistently underperformed the others, particularly when exploiting positively correlated features, indicating that overly selective attention can prevent the model from combining partial evidence. The experiment also showed the importance of measuring against a chance-level baseline, something which is not currently the norm in this area of research.

The main contribution of this project is evidence for an important condition in architectural interpretability research that an architectural constraint cannot be assumed to produce its intended representational effect simply because it is present in the model, so future interventions should first establish that the target behaviour exists before training, then verify the constraint fully survives training, and then benchmark claimed improvements against a chance-level baseline. These practices could provide a more reliable path from architectural interventions in controlled models towards real-time verifiable AI systems.

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